159102 workshop – Week 12 – Some selected solutions

1. Write a C++ program that reads in a name (could be more than one word) and an age. Then display the name and the age. Do NOT include the line `using namespace std;` Edit the program so that it runs correctly.

Solution to Q1:

```
#include <iostream>

int age;
std::string name;

int main() {
  std::cout << "Enter your name: ";
  getline(std::cin, name);
  std::cout << "Enter your age: ";
  std::cin >> age;
  std::cout << name << " is " << age;
  std::cout << " years old.\n";
}
```

2. Write a C++ program that defines three integer variables inside a namespace called RGBvalues. The program must load the values for a particular colour and then display the values.

Solution to Q2:

```
#include <iostream>
using namespace std;

namespace RGBvalues {
  int R, G, B;
}

int main() {
  cout << "Enter the Red value ";
  cin >> RGBvalues::R;
  cout << "Enter the Green value ";
  cin >> RGBvalues::G;
  cout << "Enter the Blue value ";
  cin >> RGBvalues::B;
  cout << "\nThe RGB values are (" << RGBvalues::R << ", ";
  cout << RGBvalues::G << ", " << RGBvalues::B << ")\n";
}
```

3. Modify the program in Question 2 to include the line `using namespace RGBvalues;`

Solution to Q3:

```
#include <iostream>
```

```
using namespace std;

namespace RGBvalues {
  int R, G, B;
}

using namespace RGBvalues;

int main() {
  cout << "Enter the Red value ";
  cin >> R;
  cout << "Enter the Green value ";
  cin >> G;
  cout << "Enter the Blue value ";
  cin >> B;
  cout << "\nThe RGB values are (" << R << ", ";
  cout << G << ", " << B << ")\n";
}
```

4. Write a C++ program that uses the following class along with the global array:

```
class vehicle_class {
private:
  string plate; // e.g. "ABC123"
  int year;     // e.g. 2004
public:
  void loaddata(string p, int y);
  void display();
};

vehicle_class fleet[100];
```

The program must read in details for a number of vehicles (actual number is unknown) and then display the information for every vehicle. You will need to write code for the methods.

Solution to Q4:

```
#include <iostream>
using namespace std;

class vehicle_class {
private:
  string plate; // e.g. "ABC123"
  int year;     // e.g. 2004
public:
  void loaddata(string p, int y);
  void display();
};

vehicle_class fleet[100];

// int main(int argc, char* argv[]) {
int main() {
```

```
  int i, yr, total;
  string str;
  i = 0;
  yr = 999;
  while (yr > 0) {
    cout << "Enter year of registration ";
    cin >> yr;
    if (yr > 0) {
      cout << "Enter number plate ";
      cin >> str;
      fleet[i].loaddata(str, yr);
      i++;
    }
  }
  total = i;
  cout << "\nVehicle fleet details follow:\n";
  for (i = 0; i < total; i++) {
    fleet[i].display();
  }
}

//-------------- methods for the vehicle_class ------------
void vehicle_class::loaddata(string p, int y) {
  plate = p;
  year = y;
}

void vehicle_class::display() {
  cout << "Number plate " << plate;
  cout << "   First registered in " << year << endl;
}
```

5. Modify the program in Question 4 to become a system program that uses command line arguments. All `cin` statements must be removed from the program. You will run the program from the command prompt and you will then enter something that looks like this:

```
prog1 ABC123 2001 GHY869 2015 HDT293 2018
```

Then display the information for every vehicle (as in Question 4). Note that you still do not know how many vehicles will be entered. Your program must calculate how many data entries are coming in from the command line.

Solution to Q5:

```
#include <iostream>
#include <iostream>
using namespace std;

class vehicle_class {
private:
  string plate; // e.g. "ABC123"
  int year;     // e.g. 2004
public:
```

```
  void loaddata(string p, int y);
  void display();
};

vehicle_class fleet[100];

int main(int argc, char* argv[]) {
  int i, total;   // items in the array
  int c, count;   // counting the arguments
  int j;          // index inside the year string
  int yr, len;
  string platestr, yearstr;
  count = argc - 1;  // number of arguments entered
  i = 0;
  // ignore argv[0] - this is the name of the program
  for (c = 1; c < count; c = c + 2) {
    platestr = argv[c];
    yearstr = argv[c + 1];
    yr = 0;
    len = yearstr.length();
    for (j = 0; j < len; j++) {
      yr = yr * 10 + yearstr[j] - 48;
    }
    fleet[i].loaddata(platestr, yr);
    i++;
  }
  total = i;
  cout << "\nVehicle fleet details follow:\n";
  for (i = 0; i < total; i++) {
    fleet[i].display();
  }
}

//-------------- methods for the vehicle_class ------------
void vehicle_class::loaddata(string p, int y) {
  plate = p;
  year = y;
}

void vehicle_class::display() {
  cout << "Number plate " << plate;
  cout << "   First registered in " << year << endl;
}
```

6. Do not actually run the program but predict what the following program is going to display. Note: the answer "it will display zero" is not correct.

```
#include <iostream>
using namespace std;

void thing();

int main() {
```

```
  thing();
}

void thing() {
  int value;
  cout << value << endl;
}
```

Solution to Q6:

The value inside the function is a local variable and has not been initialised. It will therefore be a random number. It is possible that it may be zero but most times it is not. REMEMBER to initialise local variables to zero when needed!